Dhana Karthikeya Ventrapragada

Berkeley, CA

+1(669)269-8955 · vdkarthikeya@berkeley.edu · Portfolio · LinkedIn · GitHub

Education

University of California, Berkeley **Dec 2027**

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Machine Learning · Data Engineering · Data Structures · Probability for Data Science · Principles & Techniques of Data Science · Structure and Interpretation of Computer Programs

Experience

Engineering Intern **May 2026 – Present**

Guang Labs *Remote*

- Shipped an **async video domain-classification service** that labels images, short clips, and long videos across **6 content domains** via **FastAPI** submit/poll endpoints backed by a **Postgres** job store and background workers.
- Built the **Gemini classifier** with **Pydantic**-validated schemas and a **confidence-gated trust layer** that auto-resamples low-trust long videos up to 10×, sampling random 10-second clips via **ffprobe/ffmpeg** and merging them through a weighted-mode rollup.
- Modularized the service into resolver, sampler, trust, and provider layers with a deterministic **fake provider** for API-key-free testing, validated by **38 passing unit/API tests**.

Data Science Intern **Jun 2026 – Present**

IDX Exchange *Remote*

- Building **end-to-end ML pipelines** on **large-scale MLS sold-property datasets** to predict California home prices, owning the full workflow from ingestion and cleaning through feature engineering, leakage prevention, and regression modeling.
- Iterating on **XGBoost** valuation models with cross-validation and feature-importance analysis; evaluating with **MdAPE, RMSE, MAE, and** R^2, and translating results into structured documentation for applied real estate analytics.

Undergraduate Course Staff (CS 61A) — UCS1 **Jun 2026 – Present**

University of California, Berkeley *Berkeley, CA*

- Serving as course staff for **CS 61A** (**250+ students**) — conducting **2 weekly exam-prep tutoring sections**, creating assignments on Gradescope, and handling grading, regrade requests, and extensions, while holding weekly office hours.

Projects

Movie Genre Classification (Multi-Label NLP) | *Python, PyTorch, SentenceTransformers*

- Built a **multi-label genre prediction system** on **42K+ MovieLens/TMDB films**, parsing nested IMDb metadata with regex, engineering log budget/revenue and temporal features, and embedding title+tagline+overview text via **MiniLM, MPNet, and e5-large-v2**.
- Benchmarked **Logistic Regression, XGBoost, and LightGBM** on 1024-dim e5 embeddings; combined a rare-genre cutoff with threshold tuning to address class imbalance, lifting **Macro F1 from 0.48 to 0.66** (Micro F1 0.68) with Logistic Regression beating tree models while training ∼12× faster.

Email Spam/Ham Classifier | *Python, scikit-learn, pandas, NumPy*

- Engineered **50+ features** from raw emails — regex counts (links, punctuation, dollar signs), HTML/subject indicators, log-scaled length features, and a curated spam/ham bag-of-words.
- Trained **L1-regularized Logistic Regression** tuned via **GridSearchCV** and validated with 10-fold cross-validation, achieving **92.1% test accuracy** on the held-out leaderboard set.

Housing Price Prediction & Assessment Bias Analysis | *Python, pandas, NumPy, scikit-learn*

- Built a **linear regression pipeline** on **200K+ Cook County Assessor (CCAO) records**, engineering log-transformed sqft, **polynomial age & geo terms, latitude×longitude interactions**, and one-hot road/garage features; filtered non-arm's-length $1 sales to align with real assessment practice.
- Achieved **test RMSE = 0.562** via 4-fold cross-validation, then quantified **systemic over- and under-estimation across price bands**, connecting design choices to documented racial inequities in property tax assessment.

Technical Skills

Languages: Python, Java, C++, SQL, Scheme
ML & Data Science: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, SentenceTransformers
Backend & Infrastructure: FastAPI, Pydantic, PostgreSQL, AWS S3, Gemini API, pytest, ffmpeg
Developer Tools: Git, GitHub, VS Code, IntelliJ, Jupyter Notebook, Google Colab, MATLAB